

Ejercicio 1 y 2

Crear base de datos

```
CREATE DATABASE bokuNoHero;
```

```
USE bokuNoHero;
```

Tabla Heroe

```
CREATE TABLE Heroe (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    nickname VARCHAR(100),  
    nombre VARCHAR(100),  
    edad INT  
);
```

Tabla Quirk

Un héroe puede tener varios quirks.

```
CREATE TABLE Quirk (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    nombre VARCHAR(100),  
    tipo VARCHAR(100),  
    hero_id INT,  
    FOREIGN KEY (hero_id)  
    REFERENCES Heroe(id)  
    ON DELETE CASCADE  
);
```

Ejercicio 3

src/

|

|— model

| |— Heroe.java

| |— Quirk.java

|

|— dao

| |— HeroeDAO.java

| |— QuirkDAO.java

|

|— database

| |— Conexion.java

|

|— main

|— Main.java

Modelo Heroe

```
package model;
```

```
import java.util.ArrayList;
```

```
public class Heroe {
```

```
    private int id;
```

```
    private String nickname;
```

```
    private String nombre;
```

```
    private int edad;
```

```
    private ArrayList<Quirk> quirks;
```

```
    public Heroe(int id,
```

```
                String nickname,
```

```
                String nombre,
```

```
        int edad) {  
    this.id = id;  
    this.nickname = nickname;  
    this.nombre = nombre;  
    this.edad = edad;  
    quirks = new ArrayList<>();  
}  
  
public int getId() {  
    return id;  
}  
  
public String getNickname() {  
    return nickname;  
}  
  
public String getNombre() {  
    return nombre;  
}  
  
public int getEdad() {  
    return edad;  
}  
  
public ArrayList<Quirk> getQuirks() {  
    return quirks;  
}  
  
public void addQuirk(Quirk q) {  
    quirks.add(q);  
}  
}
```

Modelo Quirk

```
package model;

public class Quirk {

    private int id;

    private String nombre;

    private String tipo;

    public Quirk(int id,
                  String nombre,
                  String tipo) {

        this.id = id;

        this.nombre = nombre;

        this.tipo = tipo;

    }

    public int getId() {

        return id;

    }

    public String getNombre() {

        return nombre;

    }

    public String getTipo() {

        return tipo;

    }

}
```

Ejercicio 4

```
package database;

import java.sql.Connection;
import java.sql.DriverManager;

public class Conexion {

    private static Conexion instancia;

    private Connection con;

    private Conexion() {

        try {

            con = DriverManager.getConnection(

                "jdbc:mysql://localhost/bokuNoHero",

                "root",

                "admin");

            System.out.println("Conexión correcta")

        } catch (Exception e) {

            System.out.println(e.getMessage());

        }

    }

    public static Conexion getInstance() {

        if (instancia == null) {

            instancia = new Conexion();

        }

        return instancia;

    }

    public Connection getConnection() {

        return con;

    }

}
```

Ejercicio 5

HeroeDAO

```
package dao;

import database.Conexion;

import model.Heroe;
import model.Quirk;
import java.sql.*;

public class HeroeDAO {

    private Connection con;

    public HeroeDAO() {

        con = Conexion

            .getInstance()

            .getConnection();

    }

    public int insertarHeroe(Heroe h) {

        try {

            String sql =

                "INSERT INTO Heroe(nickname,nombre,edad) VALUES(?,?,?)";

            PreparedStatement ps =

                con.prepareStatement(

                    sql,

                    Statement.RETURN_GENERATED_KEYS);

            ps.setString(1, h.getNickname());

            ps.setString(2, h.getNombre());

            ps.setInt(3, h.getEdad());

            ps.executeUpdate();

            ResultSet rs = ps.getGeneratedKeys();

            if (rs.next()) {

                int idHeroe = rs.getInt(1);
```

```

        for (Quirk q : h.getQuirks()) {
            insertarQuirk(q, idHeroe);
        }
        return idHeroe;
    }
} catch (SQLException e) {
    System.out.println(e.getMessage());
}
return 0;
}

private void insertarQuirk(Quirk q, int idHeroe)
    throws SQLException {
    String sql =
        "INSERT INTO Quirk(nombre,tip,heroe_id) VALUES(?,?,?)";
    PreparedStatement ps =
        con.prepareStatement(sql);
    ps.setString(1, q.getNombre());
    ps.setString(2, q.getTipo());
    ps.setInt(3, idHeroe);
    ps.executeUpdate();
}
}

```

Main

```

package main;

import dao.HeroeDAO;
import model.Heroe;
import model.Quirk;

public class Main {

    public static void main(String[] args) {

```

```
HeroeDAO dao = new HeroeDAO();
```

```
Heroe deku =
```

```
    new Heroe(  
        0,  
        "Deku",  
        "Izuku Midoriya",  
        16);
```

```
deku.addQuirk(  
    new Quirk(  
        0,  
        "Sin Don",  
        "Nada"));
```

```
deku.addQuirk(  
    new Quirk(  
        0,  
        "One For All",  
        "Emisor"));
```

```
deku.addQuirk(  
    new Quirk(  
        0,  
        "One For All",  
        "Emisor"));
```

```
    new Quirk(  
        0,  
        "One For All",  
        "Emisor"));
```

```
dao.insertarHeroe(deku);
```

```
Heroe mirio =
```

```
    new Heroe(  
        0,  
        "Lemillion",  
        "Mirio Togata",  
        18);
```

```
mirio.addQuirk(  
    new Quirk(  
        0,  
        "Permeabilidad",  
        "Transformación"));
```

```
    new Quirk(  
        0,  
        "Permeabilidad",  
        "Transformación"));
```



```
dao.insertarHeroe(mirio);

Heroe tomura =
    new Heroe(
        0,
        "Tomura Shigaraki",
        "Tenko Shimura",
        22);

tomura.addQuirk(
    new Quirk(
        0,
        "Deterioro",
        "Emisor"));

tomura.addQuirk(
    new Quirk(
        0,
        "All For One",
        "Emisor"));

dao.insertarHeroe(tomura);

System.out.println("Héroes insertados.");
}
}
```

Ejercicio 6

Añadir en HeroeDAO:

```
public void mostrarHeroes() {  
    try {  
        String sql =  
            "SELECT h.nickname, h.nombre, h.edad," +  
            " q.nombre AS quirk, q.tipo " +  
            "FROM Heroe h " +  
            "LEFT JOIN Quirk q " +  
            "ON h.id = q.heroe_id";  
        Statement st =  
            con.createStatement();  
        ResultSet rs =  
            st.executeQuery(sql);  
        while (rs.next()) {  
            System.out.println(  
                rs.getString("nickname") +  
                " | " +  
                rs.getString("nombre") +  
                " | Edad: " +  
                rs.getInt("edad") +  
                " | Quirk: " +  
                rs.getString("quirk") +  
                " | Tipo: " +  
                rs.getString("tipo"));  
        }  
    } catch (SQLException e) {  
        System.out.println(e.getMessage());  
    }  
}
```

```
}
```

Llamada desde Main:

```
dao.mostrarHeroes();
```

Ejercicio 7

Añadir en HeroeDAO:

```
public void edadMedia() {  
    try {  
        String sql =  
            "SELECT AVG(edad) AS media FROM Heroe";  
        Statement st =  
            con.createStatement();  
        ResultSet rs =  
            st.executeQuery(sql);  
        if (rs.next()) {  
            System.out.println(  
                "Edad media: " +  
                rs.getDouble("media"));  
        }  
    } catch (SQLException e) {  
        System.out.println(e.getMessage());  
    }  
}
```

Llamada desde Main

```
dao.edadMedia();
```